

Aix Marseille Université

Master 1 GAED Geomatics and Spatial Modeling

GIS and spatial analysis

Academic year: 2025-2026

ESTIMATION OF NOISE POLLUTION IN PERI-AIRPORT AREAS

Marignane-Vitrolles-Saint-Victoret

Mohamed SOIDRIDINE

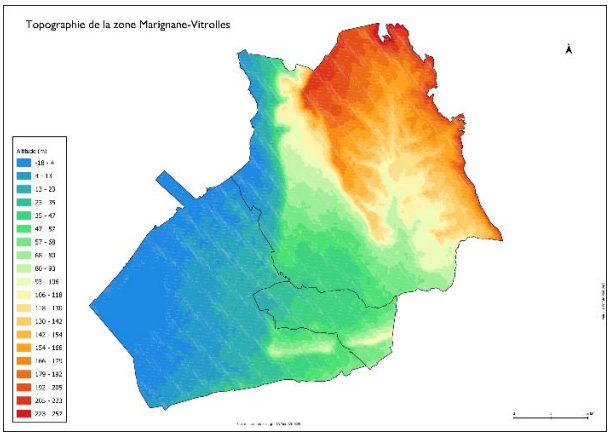
Table des matières

I. Introduction 3
I.1. A space at the crossroads of development issues and environmental obstacles 3
I.2. A territory exposed to the problem of noise pollution 3
I.3. Structure of the analytical framework 4
II. Data and methods 4
II.1. Acoustic waves 4
II.2. Acoustic indicators 4
III. Spatial analysis and mapping 5
III.1. Spatially differentiated vulnerability 5
III.2. Mapping and analysis of noise pollution 8
III.2.1. Nuisances related to airborne noise 9
III.2.2. Nuisances related to road noise 11
III.2.3. Nuisances related to railway noise 13
IV. The Environmental Noise Prevention Plan (PPBE) relating to the Marignane-Vitrolles-Saint-Victoret area 14
IV.1. Regulatory context 14
IV.2. Actions carried out in the former PPBE between 2014 and 2024 15
IV.3. The new PPBE 2024-2029 15
Conclusion 16

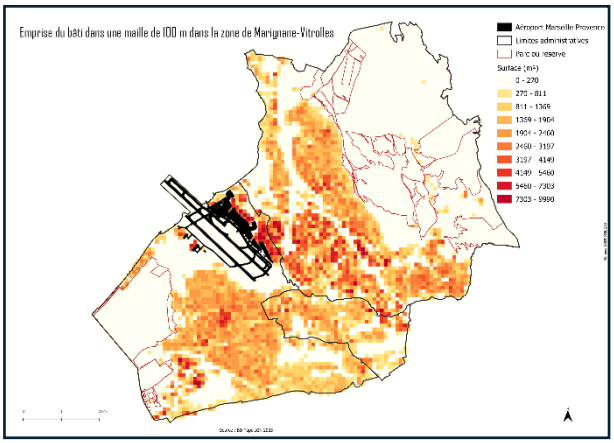
I. Introduction

I.1. A space at the crossroads of development issues and environmental obstacles

Located in the north-west of the Aix-Marseille-Provence metropolis and structured around the Etang de Berre, the municipalities of Marignane, Vitrolles and Saint-Victoret form a special peri-metropolitan territory. It concentrates major transport infrastructures on a national and international scale, including the Marseille-Provence airport located in the municipality of Marignane, whose land access necessarily passes through its neighboring municipalities, namely Vitrolles and Saint-Victoret, as well as a major industrial zone, particularly in the municipalities of Marignane and Vitrolles, and the Airbus Helicopters factory.



Topography of the study area



Footprint of the building of the study area

All of these infrastructures rest on a former marshy plain characterized by a relatively low relief that oscillates between 0 and about 257 m above sea level. It should be noted that the highest areas are in the municipality of Vitrolles (the Vitrolles plateau on the eastern part of the city culminating at an altitude of more than 200 m and offering a magnificent panorama of the Etang de Berre as well as the airport.

The Marignane-Vitrolles-Saint-Victoret territory has a dense urban fabric, which hosts a population of just over 77,000 inhabitants in a small and highly anthropogenic territory, about 76% in Saint-Victoret, 46% in Vitrolles and 72% in Marignane (Corine Land Cover 2018 data). This geographical and functional location places these municipalities at the heart of a fundamental challenge between, on the one hand, obligations to develop economically and to set up logistical connectivity, and on the other hand, the imperatives of offering a quality and living environment that is fairly acceptable to their populations.

I.2. A territory exposed to the problem of noise pollution

In this context of strong land artificialisation, the issue of noise-related nuisances is a privileged analyzer of the rather complex interferences between economic development, land use planning policy and acceptable living environment. Indeed, the study area is exposed to an accumulation of different sources of noise pollution: the presence of the two A7 and A55 motorways, the presence of the airport which creates a fairly large amount of air traffic and the rail network, in particular the high-speed railway. This accumulation of noise pollution of different origins and temporalities generates a potentially degraded acoustic layer, the study of which will be the subject of this exercise. It is therefore necessary to divide this territory into different parts and propose an estimate of the most vulnerable areas, analyze the way in which these different noise

exposures are distributed in this territory and see how the public elites try to prevent and mitigate this multi-noise exposure.

I.3. Structure of the analytical framework

This problem will be dealt with from three perspectives in order to propose an analysis that will allow a better understanding of it. Initially, an approach that can be described as health will make it possible to see the most vulnerable populations and propose a map of the most vulnerable areas. Then, in a second phase, a geographical and spatial aspect will aim to analyze and map the levels of noise exposure in space. Finally, an institutional approach will analyze the regulatory policies put in place through the Environmental Noise Prevention Plan (PPBE).

II. Data and methods

II.1. Acoustic waves

Acoustics can be defined as "*the study of sounds and their propagation in the form of longitudinal pressure waves in elastic media*" (Raphaël Ventura), at a speed that depends on the medium of propagation and the frequency. The term "sound pressure wave" is often used in a broad sense to refer to the audible frequency of humans. Different properties characterize these waves: wavelength, speed, frequency, amplitude, phase and intensity. In order to be able to measure them and set up a visualization through cartographic analysis, various instruments are used, including the sound level meter. The goal is to capture and convert the measured sound energy into another form of energy, such as electrical energy.

At the level of the Aix-Marseille-Provence metropolis, an urban monotoring system has been set up in partnership with Acoucité and AtmoSud. This initiative aims to monitor the evolution of the noise environment in areas deemed sensitive. A fleet of 10 high-precision noise measurement stations has been deployed throughout the country in order to monitor the evolution of noise pollution in the long term. These sensors convert acoustic energy into another form of energy that will make it possible to establish the spatial distribution of noise through the different acoustic indicators.

II.2. Acoustic indicators

There are various indicators for calculating the frequency of a sound, including LAeq (equivalent sound level), Ln (night level) and Lden (day, evening, night). The exercise will retain the last indicator cited (Lden) because, as its name suggests, it expresses an overall average level over the whole day (24 hours). The European Directive 2002/49/EC recommends it with the Ln index, both expressed in decibels (db(A)). It is obtained by the following formula :

$$L_{den} = 10 \log_{10} \left(\frac{12}{24} 10^{L_{day}/10} + \frac{4}{24} 10^{(L_{evening}+5)/10} + \frac{8}{24} 10^{(L_{night}+10)/10} \right)$$

With :

- Day : 7am-7pm
- Evening : 7pm-11pm
- Night : 11pm-7am

During the evening and night periods, penalties of 5dB and 10dB are introduced to reflect the annoyance of passers-by. Please note that the opening hours are for information purposes only.

It is also essential to note that the noise caused by planes is not linked to a particular day but rather to a full year, i.e. 365 days while maintaining the same logic (day-night-evening). This can pose a measurement problem, in particular because it is often not possible to keep a sensor running at the same site for 365 days.

II.3. Data collect

All the data used in this exercise to model this multi-exposure to noise was collected on the open data site of the Aix-Marseille-Provence metropolis « [MDATA] Accueil — Open Data Métropole Aix-Marseille Provence ».

Another part of this work will consist of carving out the territory in order to better observe the phenomena on a fine scale. We considered here the Iris division of the INSEE. Combined with the population census in 2022 put online in October 2025, we will have a distribution of the population by Iris, in order to see the distribution of the most vulnerable, we will consider here people aged 65 and over but also those with the lowest incomes (employees and manual workers), because, in theory, a wealthy person has more ways to have the walls of their home insulated effectively.

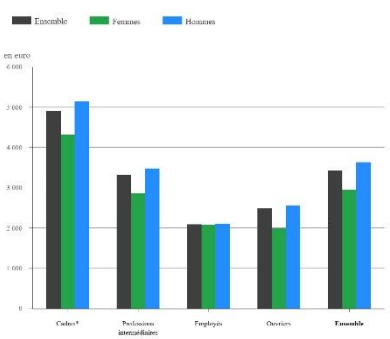
All these elements combined with different layers (administrative boundaries, buildings, etc.) will create a vulnerability index that can estimate the most vulnerable parts.

III. Spatial analysis and mapping

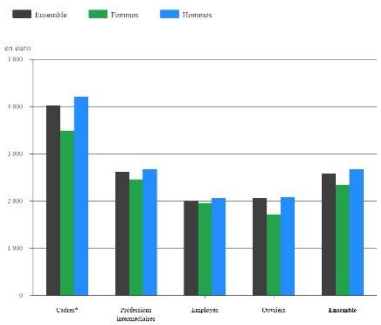
III.1. Spatially differentiated vulnerability

As stated at the beginning of the exercise, the Marignane-Vitrolles-Saint-Victoret area is highly urbanised. It is therefore understood that it is densely populated. Potentially, there are nearly 77,000 inhabitants exposed to the noise pollution that affects the study area. However, it is important to know that not all this population is exposed in the same way. It can depend on the location with regard to the sources of noise, the social status (the richer the person, the more he or she lives in a house far from nuisances or otherwise he or she has more means to install effective insulation, etc.), age (older people are much more vulnerable compared to young people for example), etc.

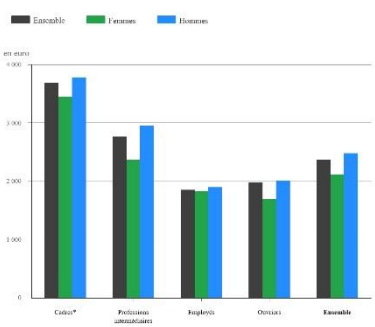
In this analysis, only age (people aged 65 and over) and socio-professional category (white-collar workers and blue-collar workers) are taken into account.



Marignane

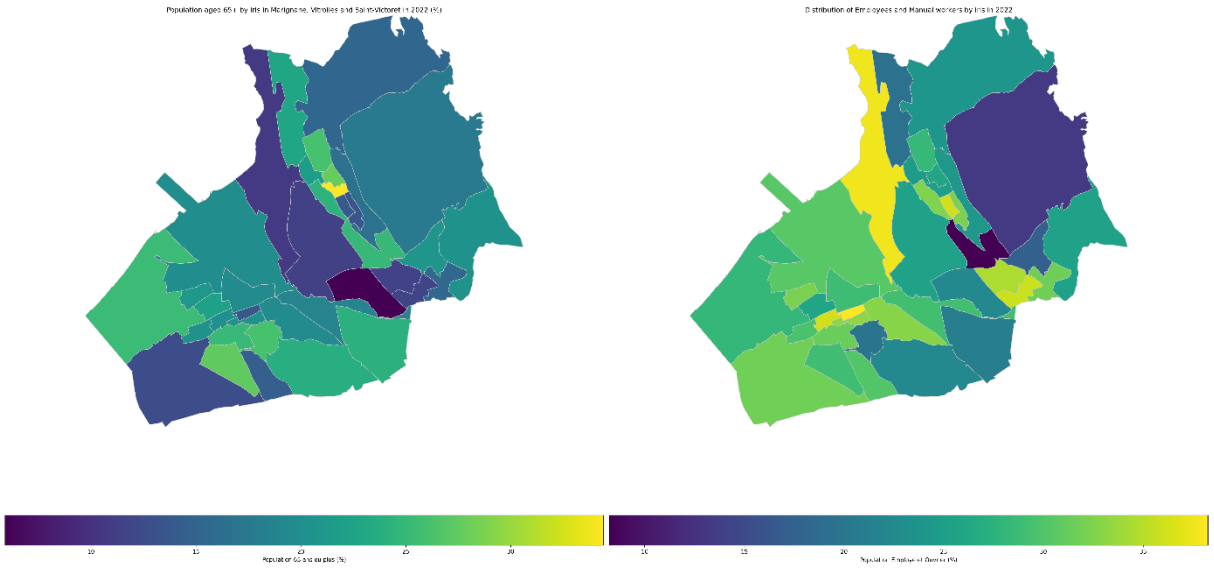


Vitrolles



Saint-Victoret

The diagrams above represent the average net salary by socio-professional category in 2023 (INSEE). We see that the people with the "lowest" monthly incomes are manual workers and white-collar workers. They will then be mapped by Iris as well as the elderly in order to better analyze their spatial distribution.



(a)Distribution of persons aged 65 and over (left) and (b)distribution of Employees and manual workers in the study area (right), Insee

Map (a) shows an unequal distribution of older people across the three municipalities. We can see that more than 30% is located in the commune of Vitrolles at the level of the Iris “Les Plantiers”, close to the historic centre. The other municipalities recorded significant results (between 15 and about 28%), particularly in the irises to the south and west of the airport (“Les Perussons-Le Moulin”, “La Filosette”, “Les Hameaux” in Saint-Victoret, etc.), which are essentially residential areas.

Map (b) also shows a very unequal distribution of this socio-professional category. With percentages of more than 35% in the municipalities of Marignane, “La Calagovière-Parc Camoin” and Vitrolles, “L'Agneau-Vignette-La Gare-Couperigne”, which is quite logical because it is essentially an industrial area (south of the Iris) and a commercial area, but it is also an area very exposed to road noise.

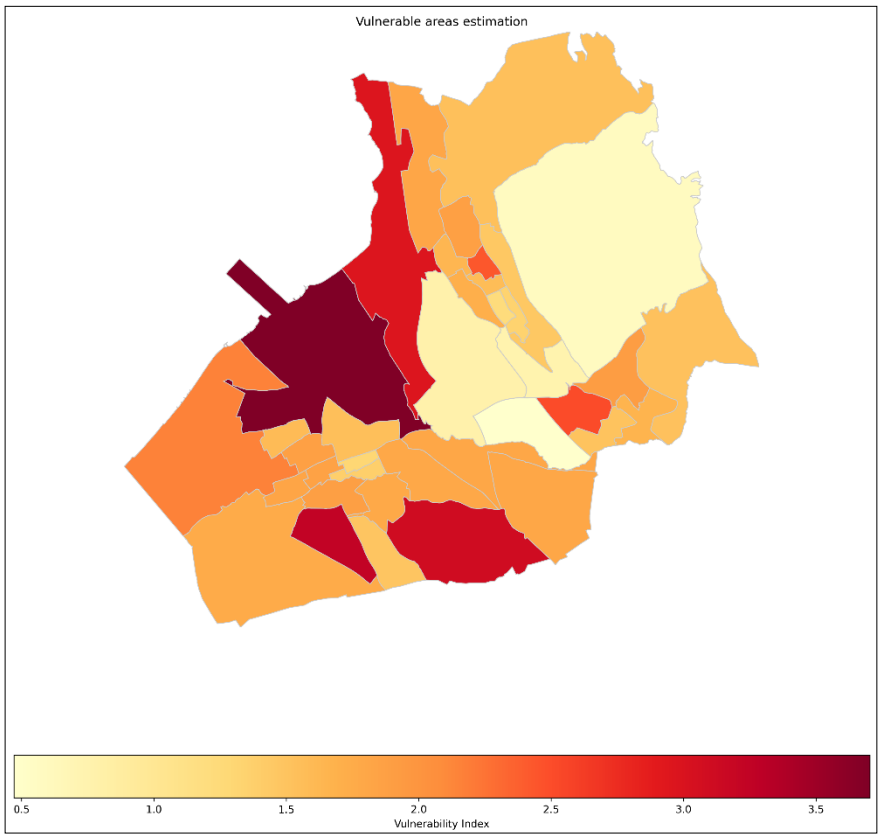
From there, the objective is to try to create a vulnerability index by combining the different available data. With programming on the Google Colab interface, the idea is to rely on proxies from the previous results (age and CSP) and the layers of administrative boundaries and buildings (BD topo and OpenStreet Map). First, after calculating the population sensitivity indicators, it is a question of calculating those of the sensitivity of the infrastructures: the density of residential buildings, grouping them by Iris, the more residential buildings an Iris contains and the more vulnerable it is, as well as the number of hospitals and clinics, they represent establishments that concentrate very vulnerable people regardless of their age. Note, we could have taken into account the schools too but being rather places frequented mainly during the day, we decided not to count them. Secondly, it is a question of identifying and counting the potential sources of noise by Iris from the layer of the building: aircraft and helicopter hangars located near the runway for example for airborne noise, train stations for railway noise, the presence of motorways or "transportation" type buildings for example for road noise.

Then, in a third step, the step is to standardize all these noise vulnerability factors by a min-max scaling. This allows all values to be put on a scale between 0 and 1.

X scaled = (x - min(x)) / max (x) – min(x))

The last step is the calculation of the noise vulnerability index varying from 0 to 3.6 approximately.

The results obtained are as follows:



Estimation du niveau de vulnérabilité aux nuisances sonores

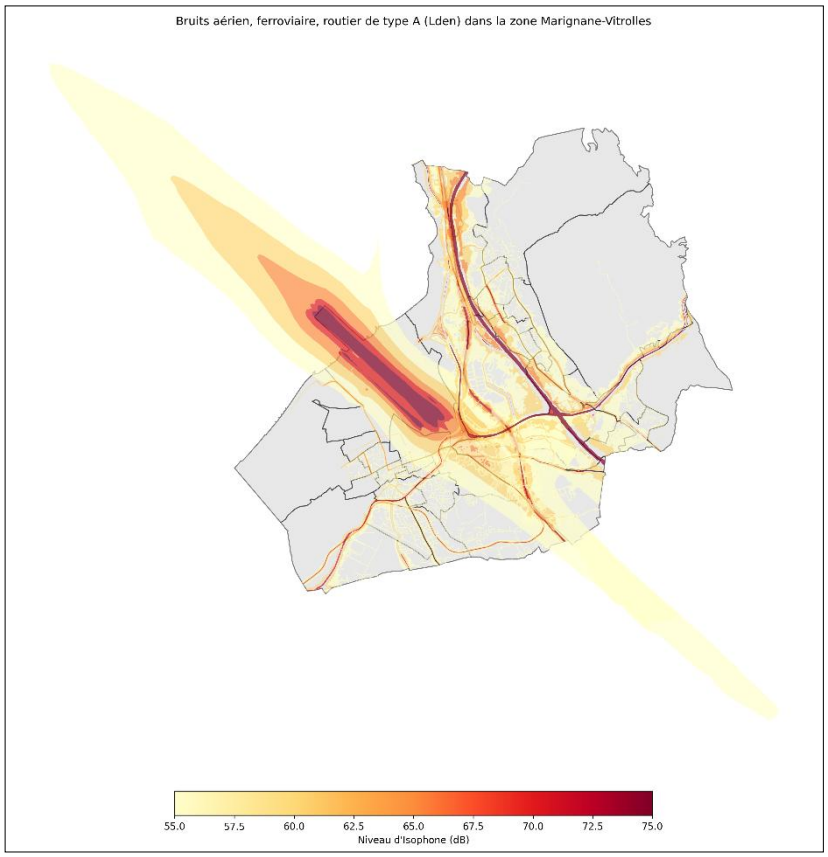
The map shows a distribution that is once again unequal at the level of the municipalities. The high values are found around the airport, with a maximum of plus 3.5 in the Iris "Estrade-Aéroport-Les Beugons", which seems normal because, due to the fact that the airport is located there, a source of major noise, it is also a residential area, especially south of the Iris, which brings together between 25 and 30% of the employees and workers, and between 18 and 20% of the elderly. Then there is the Iris "La Plaine Notre-Dame-Pas des Lanciers" to the south-east of Marignane, which is a residential area with a significant number of elderly people, "Le Carestier-Le Taulet" to the south of Marignane, an essentially residential area with a little more than 25% of the elderly, a significant presence of workers and employees, but above all there is the general clinic of the municipality. We can also mention the Iris "L'Agneau-Vignette-La Garel-Couperigne"), to the east of Vitrolles, residential to the north of the Iris, with a high rate of workers and employees, but also the presence of major roads (D13 and D20). With an index greater than 2, the areas south of the airport also have a high vulnerability. Remember that these are residential areas. As for the weakest values, they are found in the Iris in the commune of Vitrolles, "Les Plateaux Collet Rouge Montvallon", a sparsely urbanized area, "Estroublans", "L'Anjoly" and "La Bastide Blanche-Le Griffon" in the city center, which are essentially industrial and commercial areas. It is therefore understandable that not all areas or parts of the municipalities are equal, there are some much more affected than others. This prioritization will allow us to understand the rest of the exercise, which consists of mapping and modeling the different frequencies of noise in this territory.

III.2. Mapping and analysis of noise pollution

In June 2002, the European Directive on the assessment and management of noise in the environment was put in place in order to define a common approach for all member countries with the aim of avoiding, preventing or reducing nuisances related to noise exposure in the environment. One of the roles of this directive is to assess the exposure to noise of sensitive populations and equipment through Strategic Noise Mapping. The preferred sound level indicator is the Lden as mentioned earlier in this exercise. The study area, but more generally the Aix-Marseille-Provence Metropolis, are part of the territories concerned by the application of the directive.

The challenge is therefore to produce noise maps, graphic documents and data tables, and to make this data available to the population. This work is therefore based on the data available from the metropolis for the production of maps and the estimation of issues related to noise pollution. Before analysing the results, it is imperative to note that humans, from a certain audible frequency, have limits with regard to noise. But also, that a noise considered bearable (between 50 and 60 dB for example) can become harmful when it becomes repetitive or when it is constant.

Our study is limited to air, road and rail noise pollution. Here are the results from the data from the metropolis.



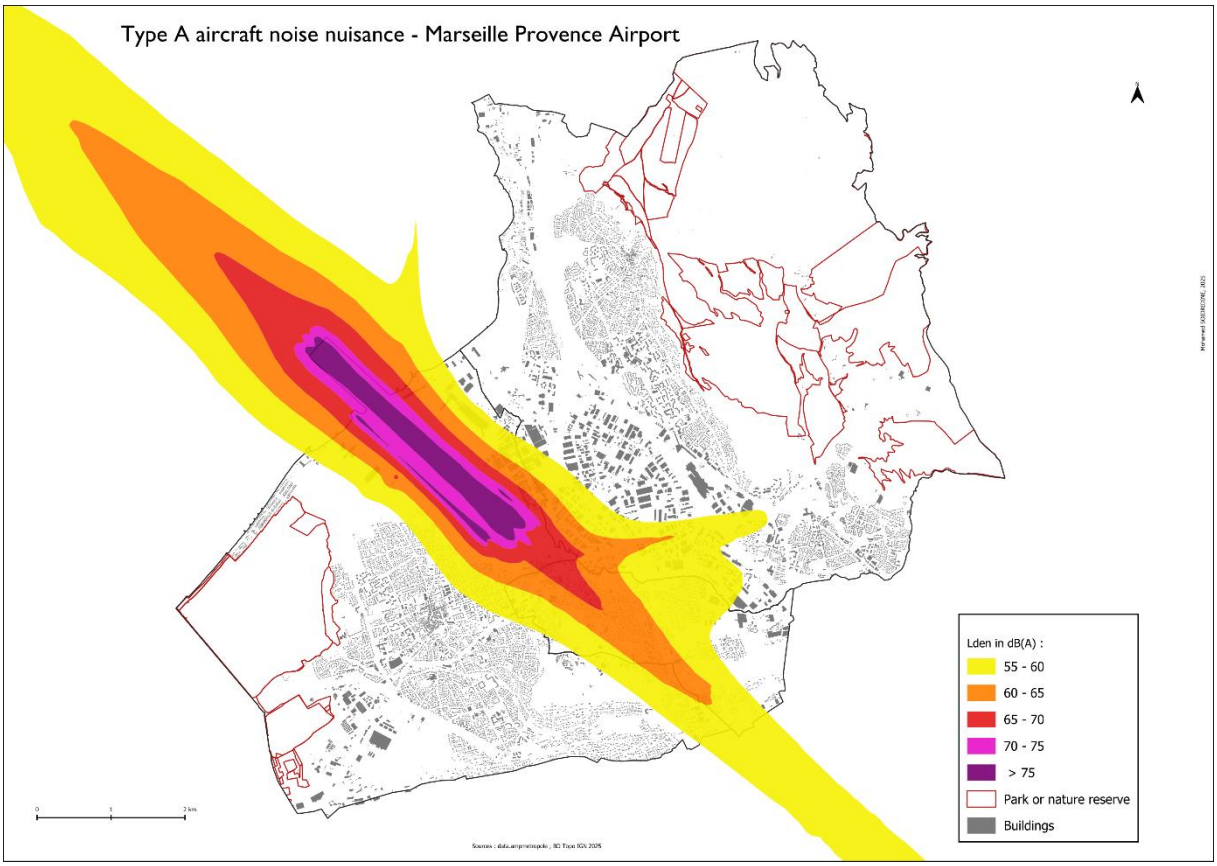
Air, road and rail noise pollution in the study area

They show a linear task with fairly high values and a fairly long and wide range. This is the measurement of airborne noise.

An important thing to note is the concentration of noise in the center of the study area. This is the area with the most interactions and activities. It represents the heart of the municipalities of Marignane and Vitrolles.

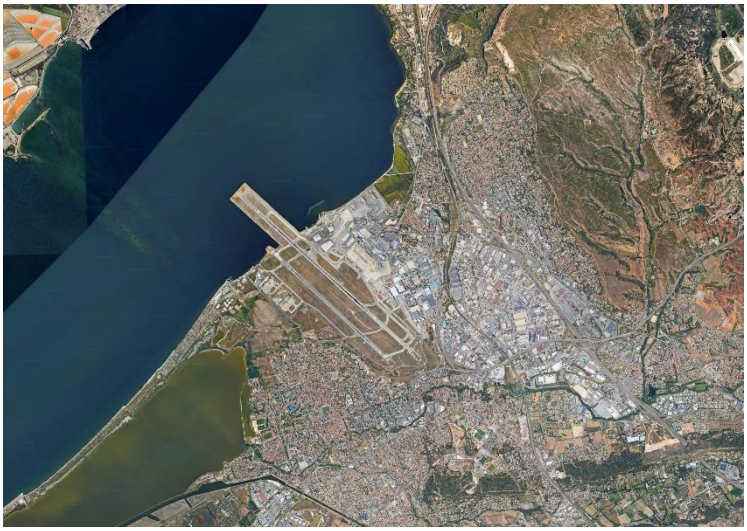
The rest of the work will consist of analysing noise by type (air, road and rail) and identifying the socio-economic issues.

III.2.1. Nuisances related to airborne noise



Assessment of the different levels of noise pollution caused by air traffic, data.ampmetropole, 2025

As its name suggests, the purpose of this map is to estimate the type A noise pollution caused by aircraft around the airport, and to highlight a hierarchy of vulnerability according to the zones. The isophone results range from 55 dB to more than 75 dB, with maximum values concentrated along the runway. Above all, it is noticeable that the scope of the task is quite long and wide, and mainly affects the municipalities of Marignane and Saint-Victoret. It also reveals the exposure of certain types of buildings, particularly residential buildings, which are directly affected by fairly high noise levels, which can exceed the critical threshold for health if the nuisances are repetitive, which is the case because the airport records up to 120 take-offs and 119 landings per day, which makes a total of about 239 flights per day (airportinfo.live). a situation that pushes the disgruntled inhabitants of these areas to stop and sometimes demand a curfew from 11 p.m. (BFMTV). According to them, the frequency of the passage of a deafening device related to planes is 10 minutes and even at night it doesn't stop. "*It's a noise that is constant, we don't have any respite. Until half past midnight, in*



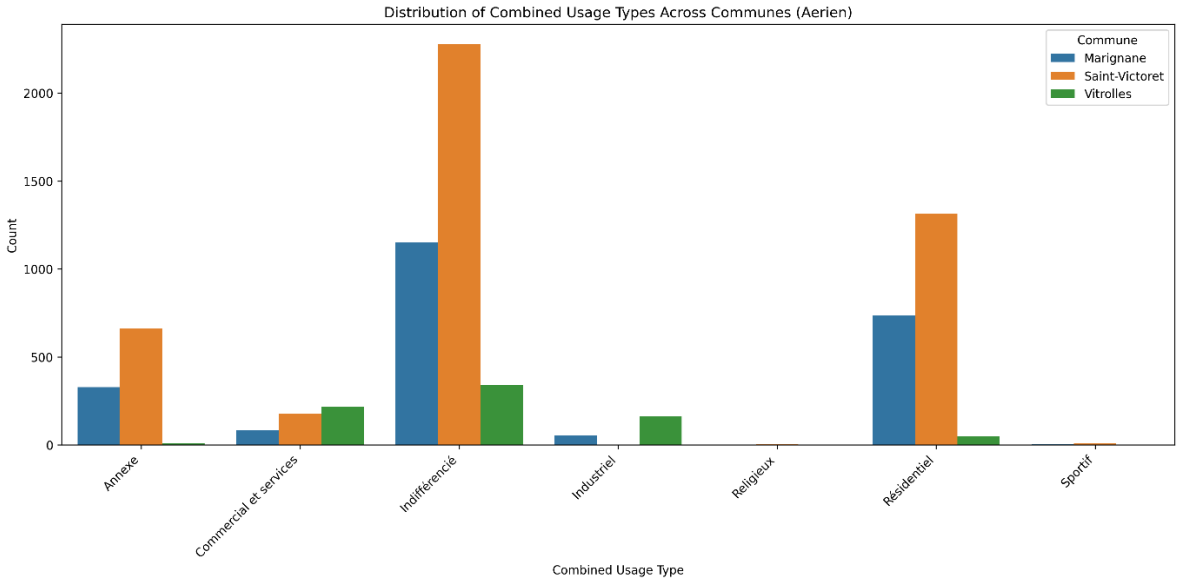
Aerial view of Marseille Provence airport in 2025, google image

the summer, we can't sleep," a resident of the areas near the airport told France Info in November 2025.



Typology of buildings exposed to air traffic nuisances around Marseille Provence airport

The image above shows the type of buildings directly exposed to noise pollution in the vicinity of the airport. We see an overrepresentation of residential, individual and collective buildings (south of the image), especially south of the airport. To the west, there is another type of building exposed to airborne noise. This is part of the industrial zone located between Marignane and Vitrolles. The stakes are therefore lower than in the residential area. The latter is one of the categories of infrastructure most affected by airborne noise, as can be seen in the graph below.



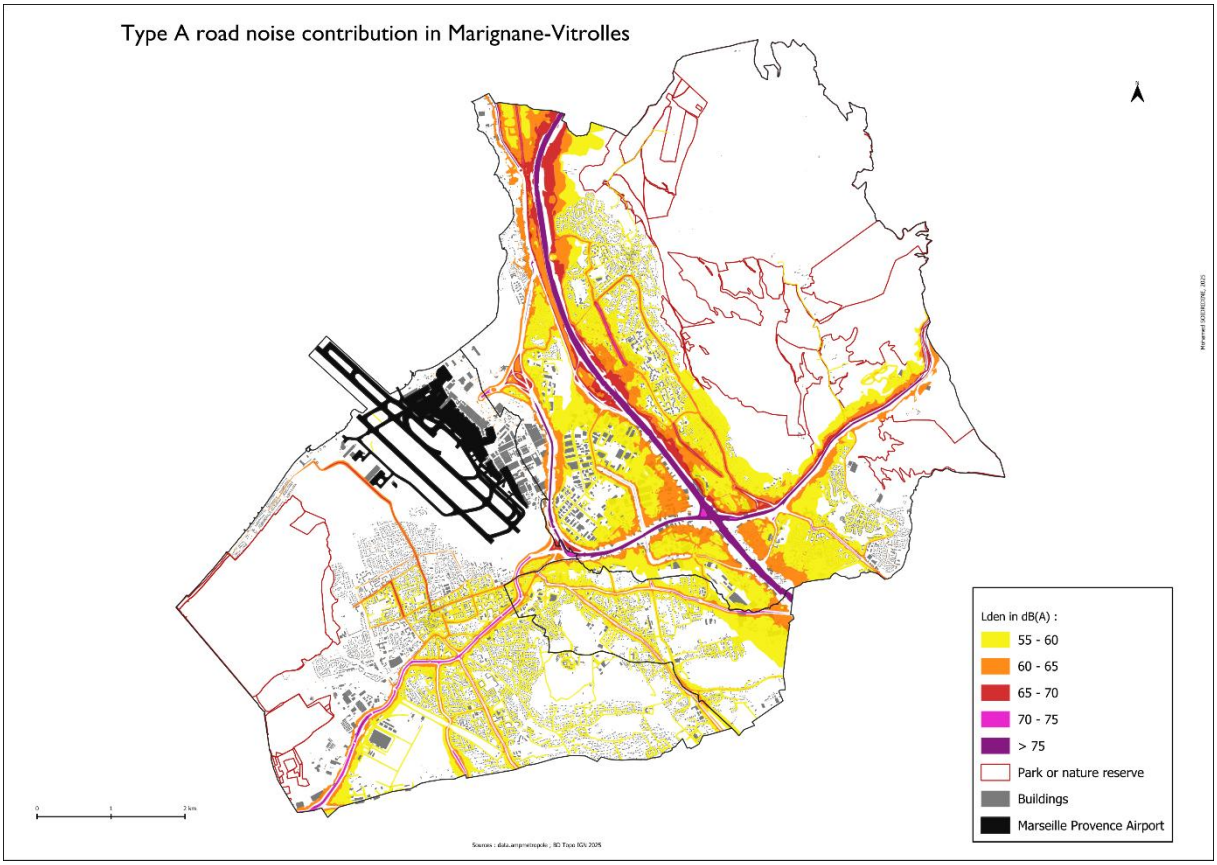
The graph highlights the distribution of uses of buildings directly affected by noise pollution in the study area. The analysis shows a predominance of undifferentiated buildings followed by

residential buildings, more precisely in the communes of Saint-Victoret and Marignane, which suggests a great exposure of these types of buildings. This is because these ships are located in the path of the take-offs and landings of the various flights.

Commercial/service and industrial uses are rather poorly represented. Again, this is due to the location of these buildings in relation to the trajectory of the planes during take-offs and landings, as we saw earlier in the analysis. The sports and religious categories are very poorly represented, which is quite logical given their small location in the vicinity of the airport.

The nuisances linked to airborne noise have a relatively linear scope, affecting almost only the buildings located at the level of the take-off and landing trajectory of aircraft, and these are essentially buildings for undifferentiated and residential use. It will be a question of seeing if it will be the same for other types of noise.

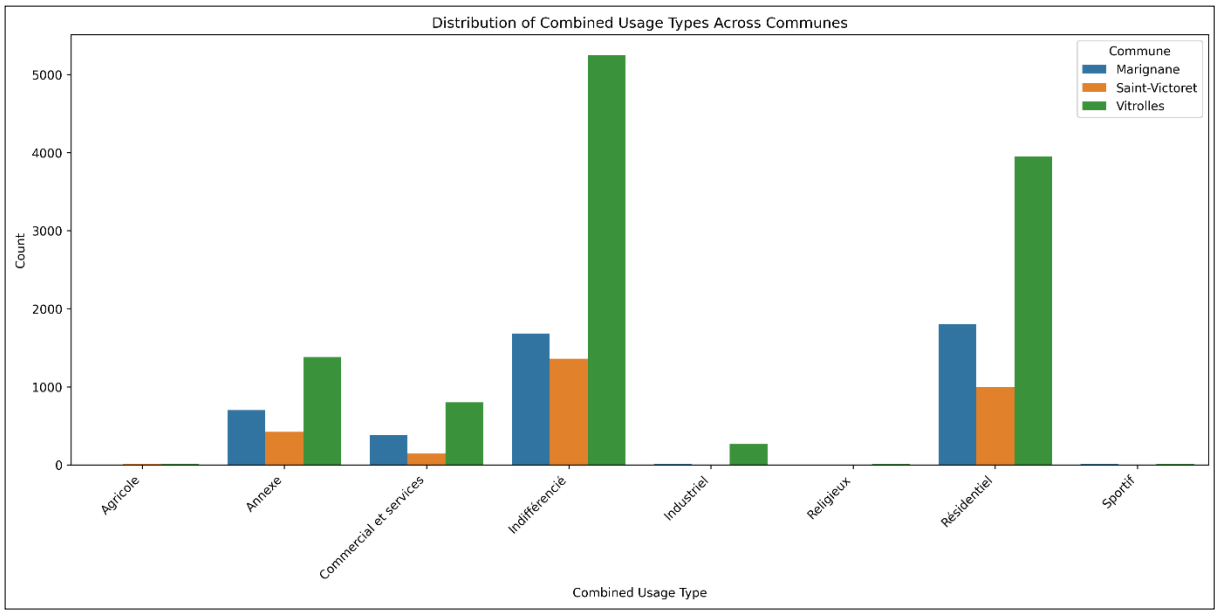
III.2.2. Nuisances related to road noise



Assessment of the different levels of noise pollution caused by road traffic, data.ampmetropole, 2025

This map will highlight the extent of noise pollution caused by road infrastructure. The values here also vary between 55 dB and 75 dB and above. The highest are concentrated along the main motorways, particularly the A7, which crosses the commune of Vitrolles from north to south, but also from the east of Vitrolles to the west of Marignane. We have a distribution of road noise, especially at levels between 55 and 60 dB, over all the urbanized areas in the three municipalities. This suggests that even remote residential areas are also exposed to noise pollution from road traffic.

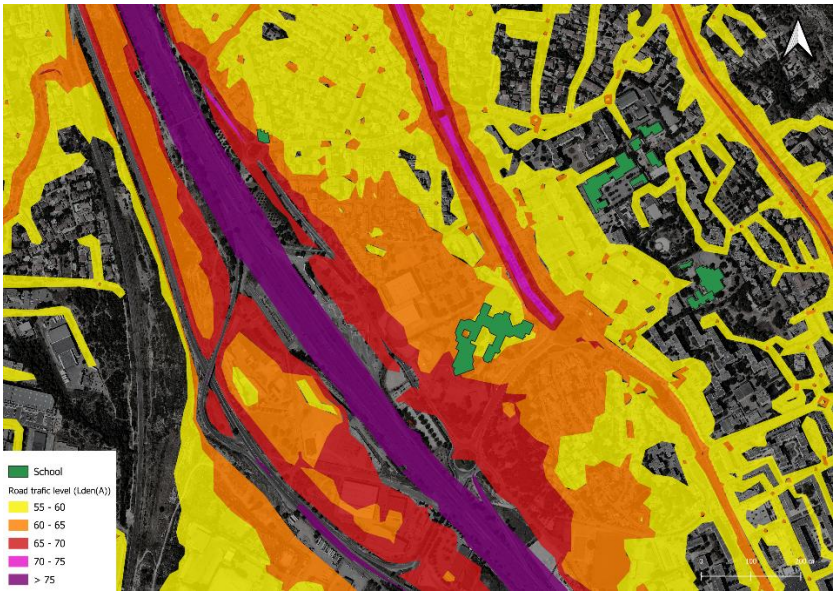
The mapping shows an almost generalized exposure to noise pollution linked to road traffic in the study area. It will therefore be interesting to see the typology of the buildings concerned.



Contrary to the previous graph, the municipality of Vitrolles is the most exposed to nuisances related to road noise. This is consistent given the location of major motorways in the Vitrolles area. Once again, we see that the most affected categories of use are residential (nearly 4000 buildings) and undifferentiated (more than 5000 buildings) in the three municipalities, even if Vitrolles remains by far the most exposed.

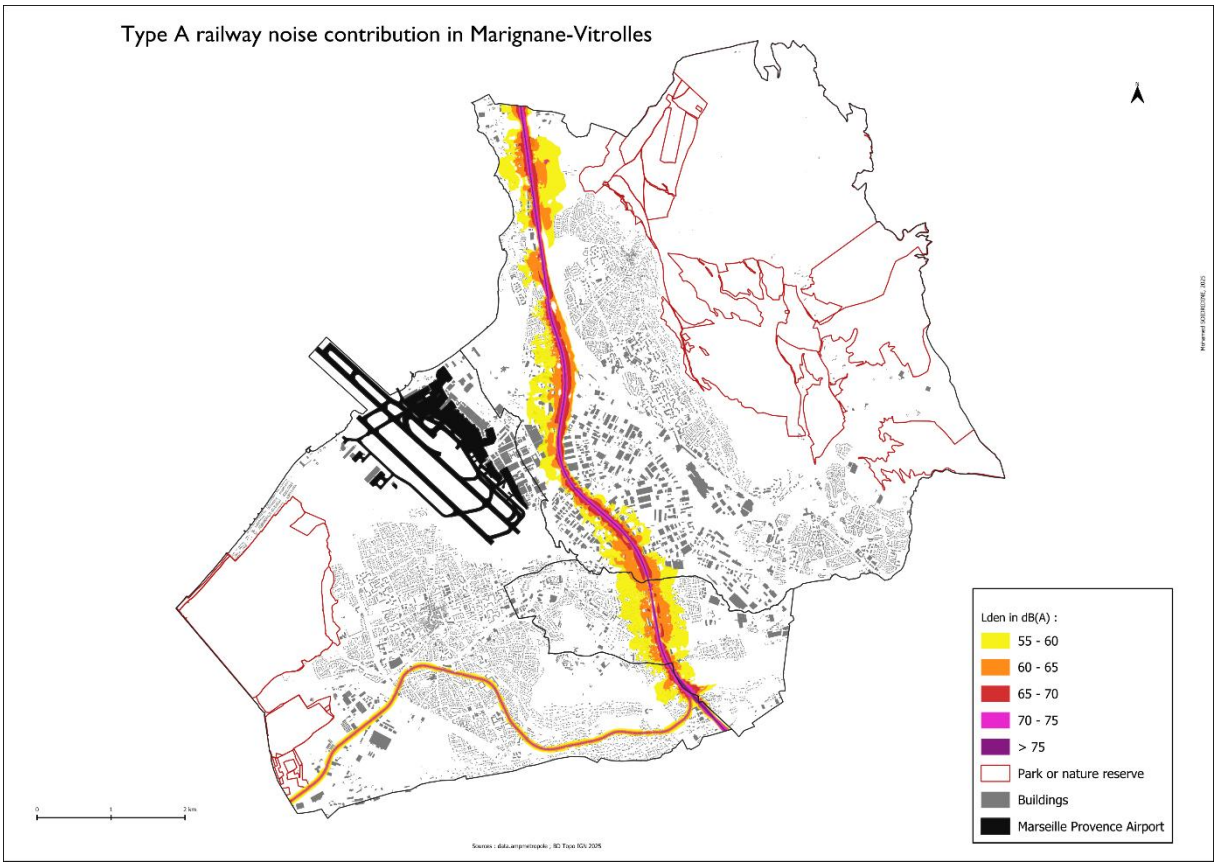
The rest of the uses have a clearly marginal exposure but they are still significant. Shops/services have a moderate number (800 buildings in Vitrolles). On the other hand, buildings for sports, industrial and religious uses are very little or not represented at all in certain areas, which suggests that they are not very well located near motorways. The distribution reflects an uneven sound exposure that particularly affects the residential areas of the municipality of Vitrolles.

However, it should be noted that residential areas are not the only vulnerable buildings exposed in our study area. If we closely analyse the various buildings located along the motorways (A7 here), we notice the presence of establishments vulnerable to noise pollution. In the image on the left, we can see the presence of schools that are very exposed to the noise pollution caused by Highway 7. This situation may raise questions about the acoustic comfort of the staff of these establishments but especially of the students. Similar situations exist almost everywhere in our study area. It will be necessary to analyse the nuisances linked to rail traffic in order to compare the results with other cases.



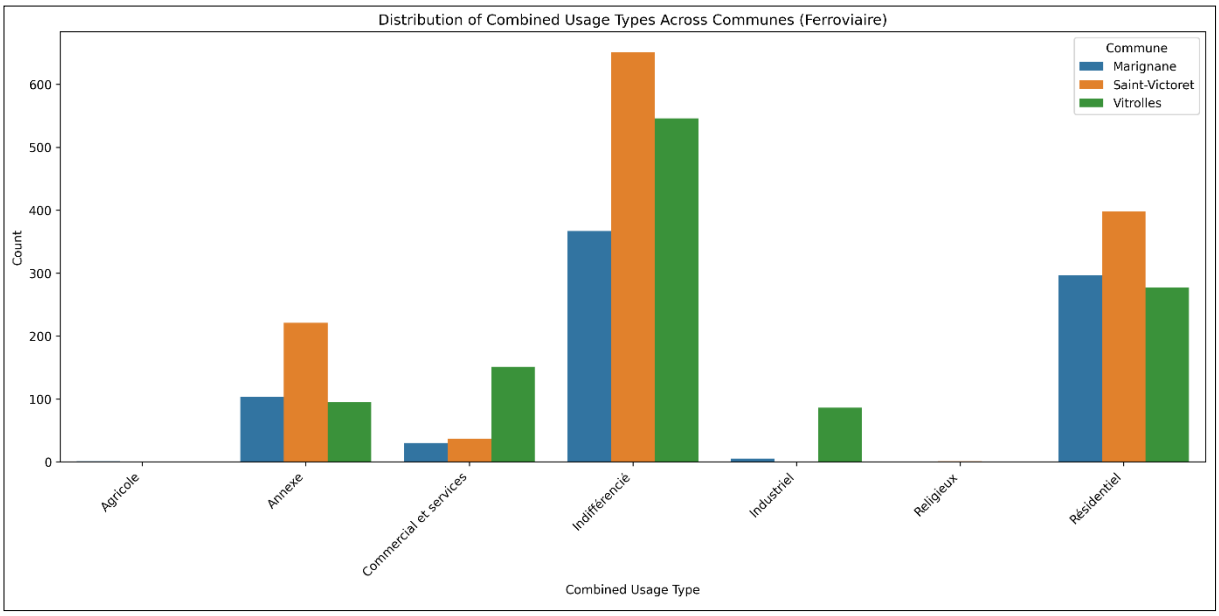
Noise level related to road traffic around the A7 in Vitrolles

III.2.3. Nuisances related to railway noise



The noise impact of railway infrastructure is totally different from that of road infrastructure. We see a line that runs from north to south, passing from Vitrolles to Saint-Victoret, forming a linear and narrow footprint but with high sound levels of the high values that are concentrated along the track. Unlike road noise, which spreads throughout the territory, railway noise has a linear range from north to south, mainly affecting buildings located near the railway line. There is also another line south of Marignane running from the east to the west. It does not have the same acoustic range as the main line, the noise does not disperse much in the surroundings. But despite this relatively short range, it crosses mainly residential areas, thus presenting itself as a source of inconvenience, especially if the frequency of passage is low. This suggests that, even if noise levels are intense, the nuisance linked to rail traffic mainly concerns a fairly small population.

The analysis of the issues related to buildings is very different from the issues of road noise that we have seen previously. The undifferentiated buildings still remain at the top of the list of the most affected but this time with values lower than the other cases analyzed, with about 645 buildings in Saint-Victoret, 550 in Vitrolles and about 370 in Marignane. Residential buildings also have an important share in this ranking, as can be seen in the graph below. 400 residential buildings in the municipality of Saint-Victoret, 300 in the municipality of Marignane and about 280 in the municipality of Vitrolles. A more balanced distribution between the three municipalities which can result in a more or less uniform crossing throughout the territory.



The rest of the categories constitute a low exposure but should not be neglected. Shops/services are very little affected, especially in Marignane (the highest concentration remains Vitrolles with 150 buildings). As for industry, it is only concerned in the territory of Vitrolles, which is consistent given the presence of the TER station located in the heart of the industrial zone of Vitrolles, only a few kilometres from the airport. Buildings related to agriculture, sports and religious practices are hardly exposed to the nuisance caused by trains.

Overall, as the study area is essentially dominated by residential buildings, exposure to noise pollution affects residential areas much more (if we decide not to take into account undifferentiated buildings) than other sectors. But in each case, the stakes are different from the others. For example, the analysis showed that exposure to railway noise appears to be much lower than that of road noise. There is therefore less at stake in the first case than in the second.

After seeing this, the next step will be to adapt a policy approach with the aim of presenting the different local urban policies put in place to mitigate noise exposure.

IV. The Environmental Noise Prevention Plan (PPBE) relating to the Marignane-Vitrolles-Saint-Victoret area

IV.1. Regulatory context

Stemming from the European Directive 2002/49/EC, the PPBE aims to define the actions put in place to prevent and mitigate noise in the environment. In our study area, it is part of a much broader framework of the PPBE of the Aix-Marseille-Provence metropolis covering a set of 92 municipalities. On 30 June 2025, a new PPBE until 2029 was adopted with the aim of "better protecting the health of inhabitants in the face of noise pollution related to transport and infrastructure", following a three-month public consultation conducted between December 2024 and February 2025. The PPBE is based on the analysis of exposure to noise pollution through a noise level mapping updated in 2021 and 2022, allowing the identification of the most sensitive areas (part (III) of this exercise).

The Metropolis is therefore placing the problems related to noise pollution at the heart of its concerns in order to prevent and protect more than 280,000 people at risk. In our study area, this

exposure is particularly increased, in particular because of the combination of several sources of noise: motorway, rail traffic and especially the proximity of the airport. The actions of the PPBE will be analysed in two parts: the old and the new PPBE.

IV.2. Actions carried out in the former PPBE between 2014 and 2024

Since its implementation, the first PPBE has made it possible to carry out concrete operations on the territory of the Metropolis. There are 11 of them as of 2014, and have made it possible to protect about 1300 homes and 3 schools. In concrete terms, these actions consisted of soundproofing buildings located near the noisiest roads, building anti-noise walls in the most exposed areas. In addition, before the extension of the ZENIBUS line to the Marignane-Vitrolles-Saint-Victoret area, mobile measurement campaigns were carried out with the aim of carrying out an inventory.

In addition to its actions, the Metropolis has put in place a preventive but above all protective policy through urban development. These measures include speed limits below 30 km/h and the deployment of bus and tram lines to reduce road traffic and noise. The Metropolis has also deployed new sensors to continuously and accurately monitor the sound environment on the scale of its territory: the Permanent Network for Acoustic Measurement.

IV.3. The new PPBE 2024-2029

This new PPBE is oriented towards three main axes, illustrating the desire of elected officials to eliminate or reduce noise pollution. Firstly, the actions consist of expanding the soundproofing programme for residential buildings and other sensitive buildings directly exposed, but also of strengthening the acoustic protection in place along the major infrastructures, in particular those of Marignane, Vitrolles and Saint-Victoret. Secondly, they consist of transforming the structuring of mobility. In concrete terms, it is a question of expanding the development plan for cycle paths (500 km), expanding public transport to a high level of services. The aim is to gradually reduce road traffic by offering less polluting alternatives. The PLU and the SCoT must also take into account the acoustic dimension. Finally, the idea is to multiply institutional cooperation and share governance. The idea is to involve everyone in this project, especially the actors directly concerned such as the SNCF or Marseille Provence Airport. Concerning the airport, this participation involves the monitoring of the Noise Exposure Plan regulating urbanization around the airport. According to the National Assembly, the noise control policy around airports is based on the concept of a "balanced approach", which consists of reducing noise through four actions: reducing noise at the source through fleet renewal, planning and managing the use of land in the surrounding area, using less noise trajectories, and restrict airport operations as a last resort.

The Metropolis will implement a transparency policy aimed at informing and warning citizens of the evolution of their acoustic environments, testifying to the level of investment of elected officials.

Conclusion

The concentration of three major sources of noise in the Marignane-Vitrolles-Saint-Victoret area places its cities among the most vulnerable in the metropolis in terms of exposure to noise pollution. A situation that mainly affects the residential areas around the sources of the noise, which strongly influences the quality of life of thousands of residents. Several methods can be used to highlight the noise generated in order to perform calculations to map the different sound levels. The results from the Metropolis' data show differentiated noise levels, ranging from the most moderate to the most harmful, in the territory.

Iris' analysis of the level of vulnerability of the different areas highlights spatial disparities within the area. The index, which cross-references noise exposure with socio-economic and demographic elements of the local population, has made it possible to identify the sectors where this noise-related pollution represents priority health and social issues. This approach could be used as a tool to prioritize the interventions of the PPBE. The latter, resulting from the European Directive 2002/49/EC, is set up with the aim of preventing and providing solutions against noise pollution. It is based on two phases. A first phase that covers the years 2014-2024 and a second phase adopted in 2025 and which runs until 2029.

However, despite the fact that all the results obtained, in particular the calculation of vulnerability, are based on sophisticated databases, it should be remembered that they are only theoretical modelling based on theories, simplifications and uncertainties. The reality of sector vulnerability or noise exposure can look different for a variety of reasons. For example, concerning the development of the vulnerability index, the modelling was based on data from Iris, but the analysis would have been further refined by creating a 500 or 1000 m grid for example in order to see better. The results would have been different, no doubt. Other characteristics could also have been taken into account when assessing the most vulnerable people, etc. The results must therefore be analyzed as orders of magnitude while taking into account their limitations.

Finally, a verification of the reality on the ground is necessary in order to compare the results obtained in order to be able to decide on the veracity of the modelling.